Efficient Dependency Resolution in IFC-aware Decentralized Programming

Steffan Christ Sølvsten and Aslan Askarov

PRiSC 2026

AARHUS UNIVERSITY

Module systems are difficult!

$\boxed{\mathtt{A}}$ $\boxed{\mathtt{B}}$

[1] Using hashes to uniquely identify an object/class is called a "*fingerprint*" in O. Arden et al. *Sharing Mobile Code Securely with Information Flow Control* (2012).

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How can one add a module cache?

How can one add a ~~module~~ *dependency* cache?

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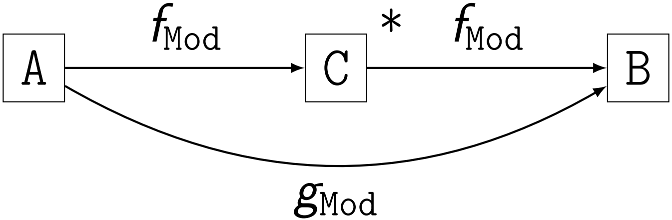

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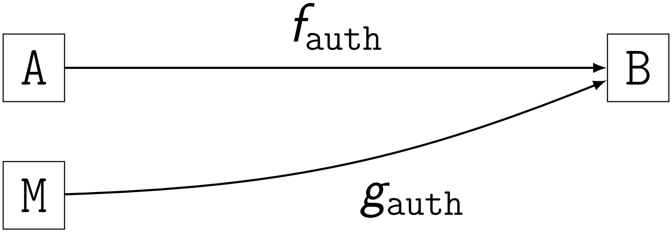

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Chain of Death[4] (Availability)

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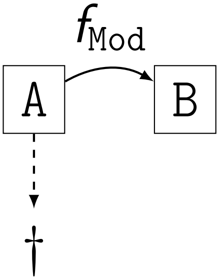

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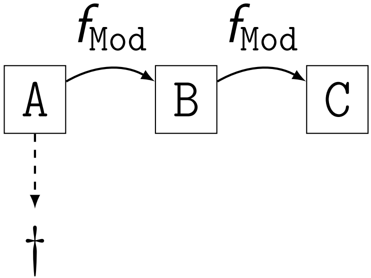

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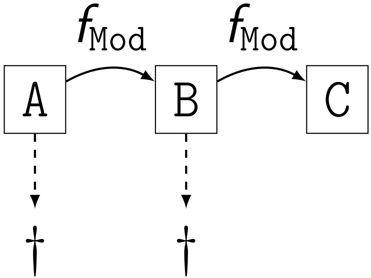

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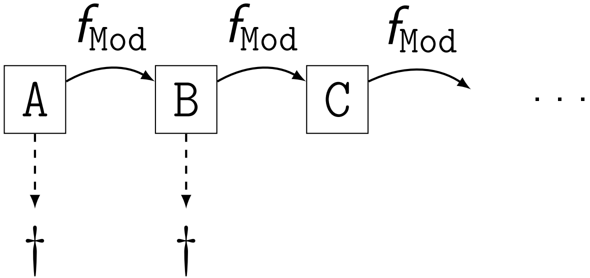

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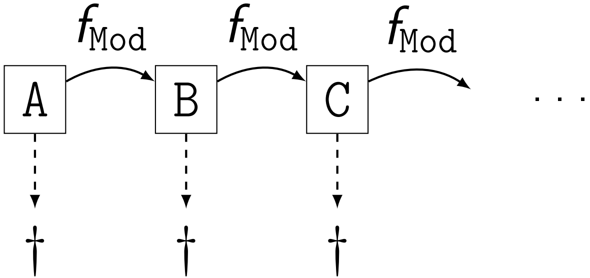

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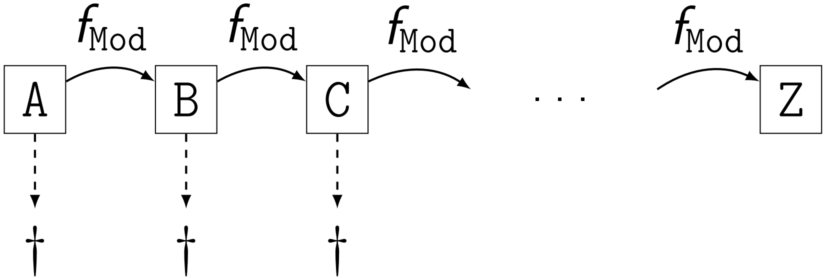

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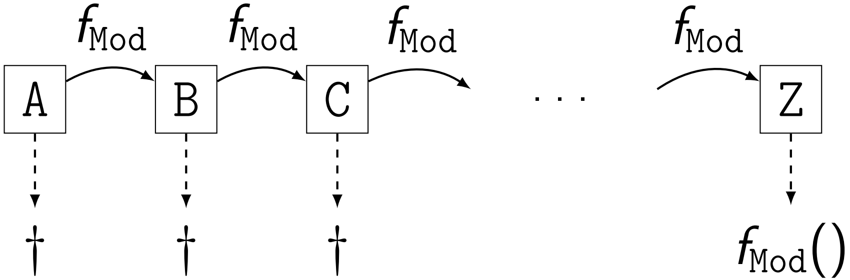

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How can one add an **IFC-aware** cache?

Troupe Programming Language

```
(* Alice.trp *)
let val msg = "Hello, Bob!"
    fun f () = print msg
in send (@bob, f)
end
```

```
(* Bob.trp *)
let fun loop () =
    receive [hn f => spawn f];
    loop ()
in loop ()
end
```

- **Decentralized Programming**
  For creation of open self-organizing distributed systems.
- **Information Flow Control**
  Ensures via a monitor that secret information is not by accident shared with nor affected by unwanted parties.
- **Dynamically Typed**
  Permits the dynamic behaviour needed by a truly open and updateable system.

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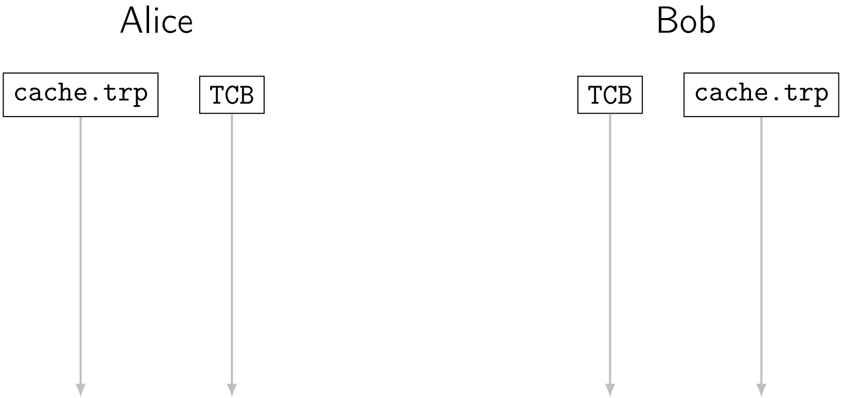
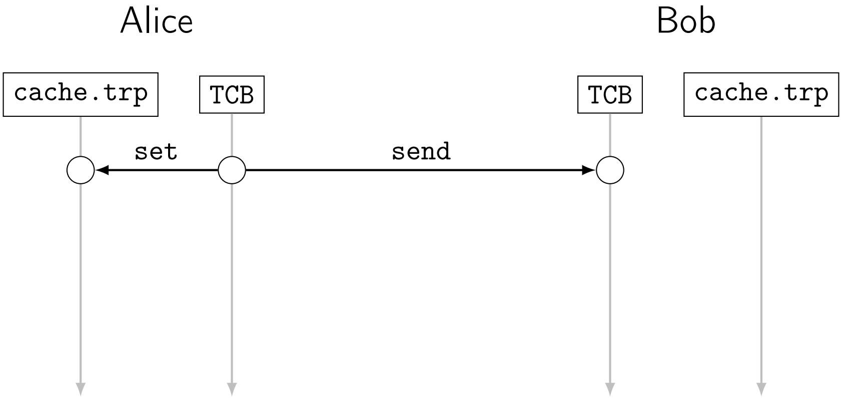
(with the minimal extension to the TCB...)

How can one add an IFC-aware cache?

(by using Troupe itself. . . ?)


Alice
Bob
cache.trp
TCB
TCB
cache.trp


Alice
Bob
cache.trp
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set
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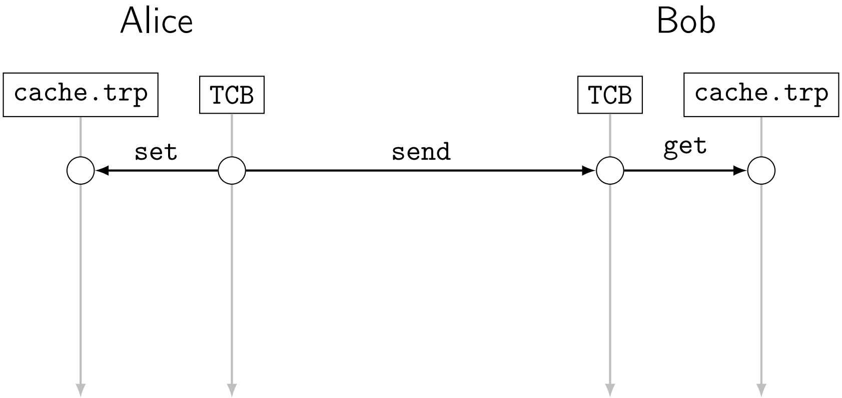
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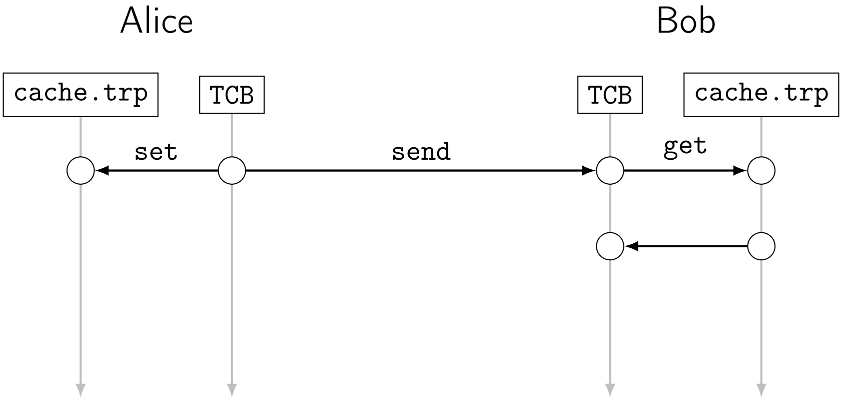
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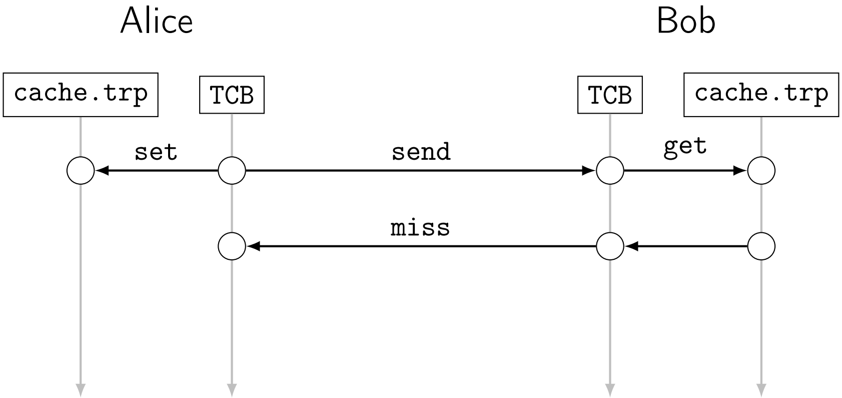
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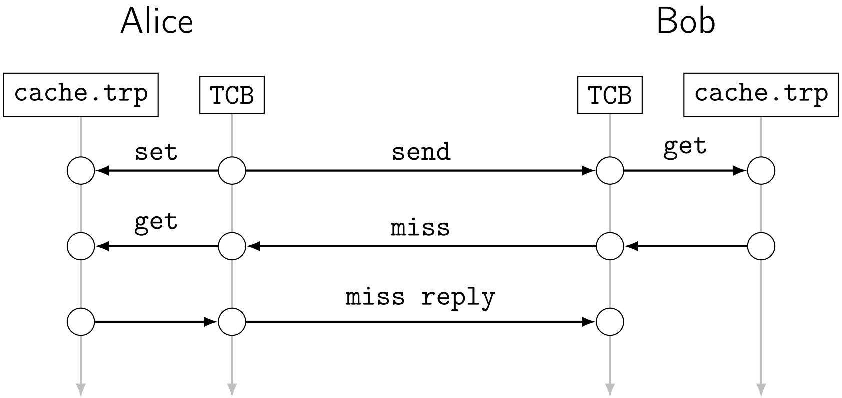
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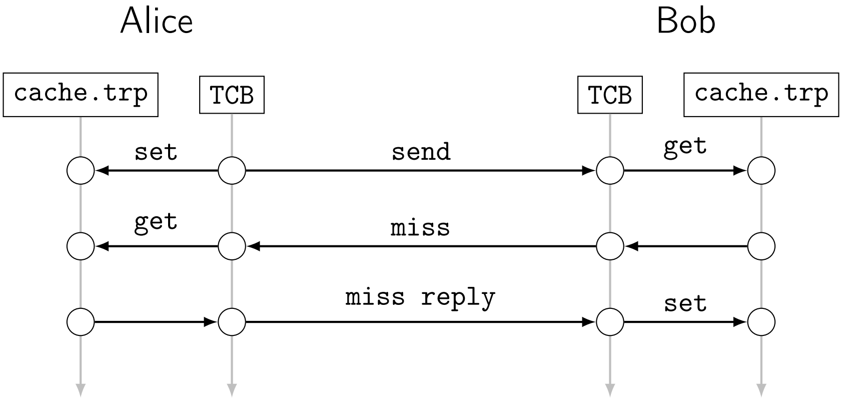
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Changes to Runtime (TCB)

- **Peer-to-peer layer:**
  Extend with `miss` and `miss reply` messages which are always sent by value.
- **Serialization:**
  Dependency resolution requires introspection of function closures.
  - During `serialize`, identify (sub)values to be referenced by hash.
  - To `deserialize`, (1) extract references (2) deserialize given (all resolved) dependencies.
- **Hash:**
  Hashing Troupe values requires introspection of function closures and source.
- **Scheduler:**
  Extend to run Troupe code from the peer-to-peer layer.

Additions in Troupe: IFC-aware `Map` / `HashMap`

$$C \xrightarrow{f_{\texttt{Mod}}@\{C\}} B \implies \text{miss} \qquad A \xrightarrow{g_{\texttt{Mod}}@\{\}} B \implies \text{miss} \qquad D \xrightarrow{h_{\texttt{Mod}}@\{D\}} B$$

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"Specification":

An IFC-aware `HashMap` is a single data structure with public and secret key-value pairs which provides a IFC-aware projections of its `insert` and `remove`[5] history:

$$\texttt{i}(x \mapsto 1)\texttt{@\{\}}, \quad \texttt{i}(x \mapsto 2)\texttt{@\{A\}}, \quad \texttt{i}(y \mapsto 3)\texttt{@\{B\}}, \quad \texttt{r}(x)\texttt{@\{B\}}, \quad \texttt{i}(y \mapsto 4)\texttt{@\{\}}, \quad \ldots$$

[5] For this particular use case, we actually only need public (`@{}`) `remove` operations.

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- Use `sandbox` builtin to guard against timing attacks. (14 LOC)

How can one add an IFC-aware cache?

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By bootstrapping the IFC-aware runtime!

Steffan Christ Sølvsten

soelvsten@cs.au.dk

ssoelvsten.github.io

Troupe

github.com/TroupeLang/troupe

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Troupe's Threat Model

Uses several security mechanisms, inspired by Jif, Fabric, and FLAM.

- Progress-sensitive Dynamic Information-flow Control
- Sandboxing
- Capabilities

Few assumptions on adversaries! But, we do ignore:

- Network Traffic Analysis. Can be avoided with via recent methods.
  (Blaabjerg and Askarov '21 & '23, Nelson, Pagnin, and Askarov '24)
- Timing-channels. Can be avoided via `sandboxing` and predictive mitigation.
  (Zhang, Askarov, Myers '10 & '12)

Additions in Troupe: caching layer (timing-attack mitigation)

Troupe provides the `sandbox`(f, t) operation to run f for t ms and either return the result, an error, or a continuation. In practice, $t = 1$ ms seems to suffice.

```
(* cache.trp : line 48 *)
val (m', t') =
    sandbox' (fn () => HashMap.insert  m auth k v) t m

(* cache.trp : line 57 *)
val (v, t') =
    sandbox' (fn () => HashMap.findAll m auth k)   t []
```

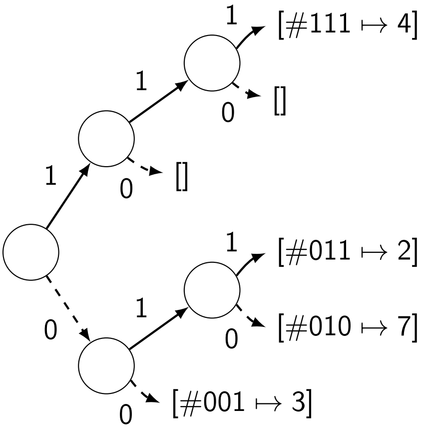
Design of the `HashMap`

Current prototype uses a Hash Array Mapped Trie (HAMT) with 200 LOC.

Only 3 declassifications per `insert`, `remove`, `find` (–24/+55 LOC):

- `hash key` and `hash bucket` to not raise `pc` and taint the entire trie.
- The `bucket` list (but not its content).

But, this fully declassifies its `size`!



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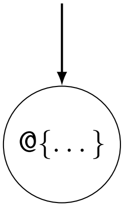

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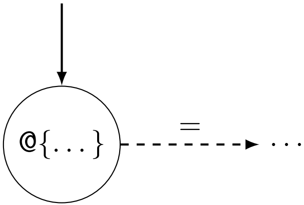

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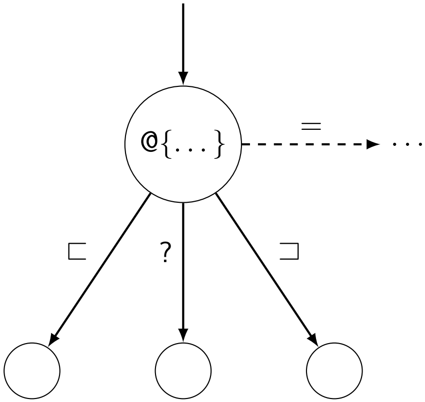

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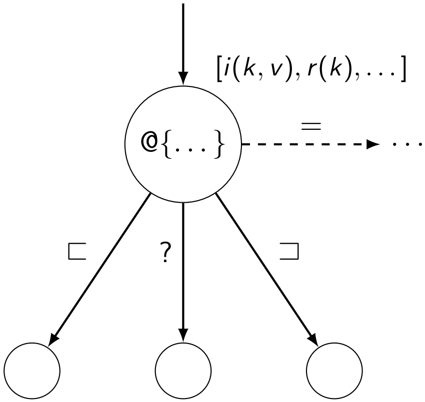

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- Use the partial ordering on information flow ($\sqsubseteq$) to compare levels.
- Use buffers to accumulate operations for "inaccessible" sub structures. This keeps `insert` and `remove` efficient.
- Use time-stamps to identify latest key-value from sub structures.

But, does this make `find` too slow?